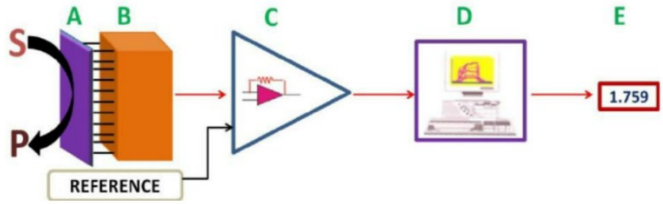
♠ EE6301 Quiz- SAMPLE ♠

Matriculation No: **Name:**

1. Given below is a diagram of the structure of a typical biosensor. What is the function of C?



a) To convert biological signals into electrical signals
b) To filter out background
c) To amplify low intensity signals
d) To serve as a bio-interface for active sensing

2. Which of the following biosensors use the movement of electrons produced during biochemical reactions?
a) Amperometric biosensor
b) Piezo-electric biosensors
c) Optical biosensors
d) ELISA biosensors

3. Which of the following statements is **not** true for biosensors?
a) Biosensors can convert a biological signal into an electrical signal
b) Biosensors are used to determine the concentration of substances of biological interest even where they do not utilize a biological system directly
c) Linearity of the response should be higher the better
d) Evanescent field is used for detection when light penetrates from lower to higher refractive index

4. Which of the following statements is FALSE?
(a) Oxidation and reduction half-reactions occur at electrodes in electrochemical cells.
(b) All electrochemical reactions involve the transfer of electrons.
(c) Reduction occurs at the cathode.
(d) Oxidation occurs at the anode.
(e) All voltaic (galvanic) cells involve the use of electricity to initiate nonspontaneous chemical reactions.

5. A diffraction grating has a first-order resolving power of 1200. What is the spectral resolution centered at 630 nm?

a) 5.25 nm
b) 1.90 nm
c) 0.52 nm
d) 0.19 nm

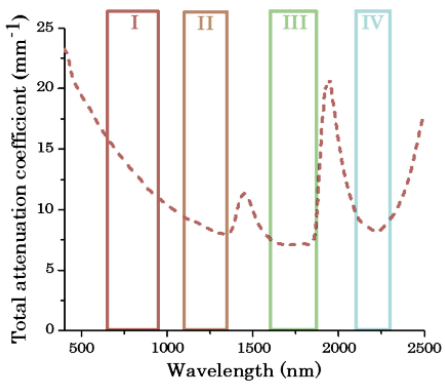
6. A human-machine interface consists of an amperometric electrochemical wearable sensor which can detect chemicals in sweat. As a proof-of-concept, below shows the sensor part for detecting glucose ($C_6H_{12}O_6$, molecular weight= 180.00; 1 mole= $6x10^{23}$).
$C_6H_{12}O_6 + 6H2O \rightarrow 6CO_2 + 24\ H^+ + 24e^-$

   How many electrons will be released in the current flow after oxidation of glucose in 0.36 mL of sweat? Assume concentration of glucose is 1.0 g/L.

a) 2.88*E19 electrons
b) 4.88*E5 electrons
c) 4.88*E16 electrons
d) 2.88*E17 electrons
e) 1.44*E19 electrons

7. Which statement best describes *stokes shift* in fluorescence process?
   A. Stokes shift is the movement of electrons from the lowest vibrational level of the first higher electronic state to any vibrational level of the lowest electronic state.
   B. Stokes shift is the change in wavelength to a shorter wavelength in fluorescence band than the excitation light due to energy loss.
   C. Stokes shift is the change in wavelength to longer wavelength in fluorescence band than the excitation light due to energy loss.

8. From the table we know that the total attenuation coefficient (absorption coefficient) differs from different wavelengths in tissue. I, II, III, IV, represents 4 optical windows for bioimaging. What is the best window to expect the highest signal? Why?
   A. Window I, lower Hb and H2O absorption
   B. Window III, lower Hb and H2O absorption
   C. Window I, higher Hb and H2O absorption
   D. Window III, higher Hb and H2O absorption



9. Which is the advantage of ultrasound imaging compared with optical imaging?
   A. Larger penetration depth
   B. Higher resolution
   C. Cross-sectional imaging mode
   D. Ability to measure distance

10. Which sentence about Raman is not correct?
   A. Photons are mostly absorbed in Raman process to generate scattering
   B. Raman shift of 5000 cm^{-1} is within the infrared region.
   C. Raman sensors require the molecule to be polarizable
   D. Raman scattering is the scattering interaction between light and particles